

Where Does the Loss Come From?

Disentangling Token Pruning and Quantization in Compressed Vision–Language Document Retrieval

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Abstract—Vision–language retrievers such as ColPali remove optical character recognition from the document search pipeline by embedding rendered page images directly, but they replace one dense vector per page with roughly a thousand patch vectors, inflating the index by two to three orders of magnitude. The standard remedy is to prune visual tokens and to quantize the survivors, and systems that do both report large end-to-end compression figures. Those figures do not say which of the two stages the retrieval quality was actually paid to. We build OptiVision RAG, a model-agnostic compression layer that applies pixel-space saliency pruning, embedding-space redundancy collapsing and one-bit quantization to any ColBERT-style visual retriever, and we run a controlled ablation in which the corpus is encoded once and every configuration replays over identical cached vectors. We run that ablation twice: on a 60-page generated corpus with a 256M-parameter encoder, and on four ViDoRe test splits (1,780 pages, 1,325 queries) with the reference ColPali-v1.3 encoder. The attribution does not transfer between them. Under the small encoder the two stages are sharply asymmetric: both pruning stages together give a $3.5\times$ token reduction for 3.9% of nDCG@5, while one-bit quantization alone costs 12.1% without discarding a single token and drives Kendall τ to 0.585. Under the reference encoder on real benchmark pages that asymmetry disappears: one-bit quantization costs between nothing and 3.7%, and the full pipeline reaches 53–60 $\times$ compression at 94.6–103.4% retention. What fails instead is the pruning premise. Real pages are dense, so both pruning stages together yield only 1.7–1.9 $\times$ rather than 3.5 $\times$, and a token-budget sweep that was flat under the small encoder falls to 66.7% retention at a 10% budget. A third experiment holds the corpus fixed and swaps only the encoder, and a per-query analysis of the encode caches says what moved. The corpus governs what pruning buys (4.2 $\times$ to 1.9 $\times$ under one encoder). The codec’s cost follows neither encoder nor corpus as such but the queries’ decision margins: a sign code perturbs page scores by about 3% under either encoder, and it flips exactly the queries whose float-precision margin over the best competitor is smaller than that — none above twice the noise, on eight caches. The generated corpus decides 60 of its 72 queries by three digits at a median margin of 0.05%, which is why the small encoder looks fragile; the reference encoder cannot read those digits at 448px and never wins the queries it is credited with not losing. On dense pages, finally, selecting tokens by embedding coverage beats pixel saliency below a 50% budget, and no selector we tried beats uniformly random probes. We conclude that per-stage attribution is necessary, and that it is a property of the queries and the corpus rather than of the compression layer — so a compression ratio for multi-vector retrieval should be published together with its attribution and

the setting the attribution was measured in.

Index Terms—document retrieval, vision–language models, late interaction, token pruning, binary quantization, index compression, ColPali

I. Introduction

Retrieval over scanned and born-digital documents has traditionally been staged: optical character recognition (OCR) extracts text, a layout parser recovers structure, a text encoder produces embeddings, and a vector database serves them. Each stage is a source of error, and the first is the worst — OCR degrades on scans, handwriting, stamps and dense tables, and the layout, figures and formatting that a human reader uses to locate information are discarded before the retriever ever sees them.

Vision–language retrievers avoid the pipeline entirely. ColPali [1] renders each page as an image, splits it into a grid of patches, and embeds every patch with a vision–language backbone, scoring a query against a page with the late-interaction MaxSim operator inherited from ColBERT [2]. No text is ever extracted, and the approach is state of the art on the ViDoRe benchmark.

The cost is the index. A page is no longer one vector but roughly a thousand patch vectors of 128 dimensions each. At float32 this is approximately 448 KB per page, so a one-million-page archive requires an index on the order of 448 GB, which must remain resident for search to be fast. The accuracy is affordable; the memory is not.

Two families of remedy exist and are usually applied together. Token reduction removes patch vectors before they are stored, on the observation that a document page is mostly blank paper; it descends from work on accelerating vision transformers, such as DynamicViT [6] and token merging [7]. Quantization shrinks each surviving vector, from product quantization [4] to the one-bit hashing used in dense passage retrieval [5]. Combining them multiplies the savings, and a system that prunes to a quarter of the tokens and stores the rest as one-bit codes can honestly advertise a compression ratio above 100 $\times$.

That headline number is also where the reporting usually stops. It says how small the index became and how much retrieval quality was lost in total, but not

which stage the quality was spent on. The distinction is not academic: the two stages have entirely different cost structures and entirely different upgrade paths. If most of the loss belongs to pruning, the productive research direction is a better saliency model. If most of it belongs to the codec, pruning harder is nearly free and the effort belongs in the quantizer instead. A single end-to-end figure cannot distinguish these cases, and the two prescriptions are opposites.

This paper performs that attribution. Our contributions are:

- 1) A model-agnostic compression layer. OptiVision RAG applies pixel-space saliency pruning, embedding-space redundancy collapsing and configurable quantization to the patch embeddings of any ColBERT-style visual retriever, leaving the encoder unmodified. Pruning decisions are made from transparent pixel statistics rather than a learned model, so the behaviour is interpretable and the operating point is a dial rather than a retrained network.
- 2) A controlled per-stage ablation. The corpus is encoded once and every configuration replays over identical cached vectors under exact brute-force MaxSim, so a change in the metric can be caused only by the compression setting. Alongside absolute retrieval metrics we report Kendall τ against the uncompressed baseline’s own ranking, which isolates whether compression changed the retriever’s mind independently of whether the retriever was right.
- 3) The finding that the attribution belongs to the setting, not to the layer. Under a 256M encoder on generated pages the two stages are sharply asymmetric, and the one-bit codec accounts for essentially all of the loss. Repeating the identical ablation with the reference ColPali encoder on four ViDoRe splits reverses this: the codec becomes nearly free, and what fails is the saliency premise that a document page is mostly blank paper. We report both regimes, give the operating points that follow in each, and argue that a compression ratio for multi-vector retrieval should be published with a per-stage attribution and the encoder and corpus it was measured on — because the attribution, not merely its magnitude, is what changes.
- 4) The mechanism, and a control for token selection. A per-query analysis over eight encode caches locates the one-bit codec’s cost in the queries’ decision margins rather than in the encoder: the codec’s score noise is the same under both encoders, and it flips exactly the queries decided by less than that noise. A matched-budget study on two dense ViDoRe splits shows that selecting tokens by embedding coverage beats pixel saliency, and that uniformly random probes do as well as a designed selector — the control any future selector has to clear.

II. Related Work

A. Late-interaction and visual retrieval

ColBERT [2] introduced late interaction: rather than compressing a document into one vector, it retains one vector per token and scores a query by summing, over query tokens, the maximum similarity against any document token. ColBERTv2 [3] reduced the resulting storage with residual compression over centroids. ColPali [1] carried the mechanism into the visual domain, treating image patches as the document tokens, and Document Screenshot Embedding [9] showed that page screenshots are a viable universal document representation. These systems establish the accuracy that motivates our work and the index size that motivates compressing it.

B. Token reduction

DynamicViT [6] learns to discard uninformative tokens during inference and token merging [7] combines similar ones, both aimed at throughput for classification backbones. The objective differs from ours in a way that matters: a classifier needs only its pooled representation to survive, whereas a late-interaction retriever needs the surviving vectors to preserve a ranking over pages. A token that never wins a MaxSim maximum is free to remove; a token that decides a close ordering is not, and classification accuracy cannot detect the difference.

C. Quantization for retrieval

Product quantization [4] and binary hashing [5] are standard for compressing dense embeddings, and modern vector databases expose both alongside approximate nearest-neighbour structures such as HNSW [8]; Qdrant [11] additionally supports multi-vector MaxSim comparators with binary quantization and rescoring. These techniques are typically evaluated on single-vector retrieval, where each document contributes one vector and quantization error affects one score. Under MaxSim the error enters through a maximum over hundreds of vectors per page, which is a different statistical regime, and it is applied uniformly with no notion of which tokens matter.

D. Positioning

The individual components we use are not novel, and we do not claim them. What is missing from the literature is a controlled attribution of quality loss between pruning and quantization when both are applied to the same multi-vector visual index. Systems report the product of the two savings; we report the ratio of the two costs.

III. Method

A. Overview

OptiVision RAG sits between an unmodified visual encoder and the vector index (Fig. 1). A page is rendered to an image and encoded into $n \approx 875$ patch vectors of dimension $d = 128$. A saliency map computed directly from the page pixels selects which grid cells survive;

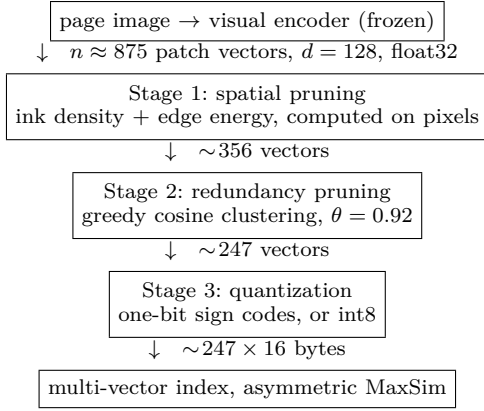


Fig. 1. The compression layer. Vector counts are the measured means over the evaluation corpus. Only the three middle stages are ours; the encoder and the index are used as published.

a redundancy pass in embedding space collapses near-duplicate survivors; a quantizer compresses what remains; and the result is stored as a multi-vector page entry scored by MaxSim. The encoder is never fine-tuned and never sees the compression, which is what makes the layer transferable across ColPali, ColQwen2 and ColSmol checkpoints.

B. Stage 1: pixel-space saliency pruning

The premise that a page is mostly paper can be tested on the pixels, before the model runs and at negligible cost. For each cell of the patch grid we compute two signals on the greyscale page. Ink density is the mean positive deviation below the paper level, where the paper level is estimated as the 95th percentile of intensity rather than a fixed white point, so that yellowed or photocopied scans are not read as ink everywhere. Edge energy is the mean absolute first difference in both axes, which responds to glyph strokes, rules, table lines and stamp boundaries. Both are normalised by a high percentile rather than the maximum, so a single dark speck — dust, a punch hole, a staple shadow — cannot squash the rest of the page toward zero. The saliency of cell (r, c) is

$$s_{rc} = \frac{w_{\text{ink}} \hat{I}_{rc} + w_{\text{edge}} \hat{E}_{rc}}{w_{\text{ink}} + w_{\text{edge}}}, \quad (1)$$

with $w_{\text{ink}} = 0.6$ and $w_{\text{edge}} = 0.4$. Ink alone drops faint handwritten annotation; edges alone retain scanner noise on an empty margin; the weighted sum is robust to both and costs microseconds. Cells with $s_{rc} > \tau_{\text{blank}} = 0.02$ are kept, and the resulting mask is dilated by one cell in 4-connectivity, because glyphs rarely stop at a patch boundary and the ascender of a heading routinely falls into a neighbouring cell whose own density is below threshold. A floor of 8 cells guarantees that a nearly empty page remains retrievable. An alternative mode keeps the top k cells by saliency, which fixes the per-page budget — this is the mode used for the sweep in Section V-B, since a



Fig. 2. Spatial pruning on a page from the evaluation corpus. Left: the rendered page. Centre: the pixel-space saliency map. Right: the retained patch mask, green for kept and red for discarded — 233 of 768 grid patches (30%) survive, and after redundancy collapsing 233 of 875 vectors (27%) are stored. The retained region tracks the text block, and the margins, which occupy most of the page, cost nothing.

fixed budget is what index-size guarantees require. Fig. 2 shows the two modes’ effect on a representative page.

C. Stage 2: embedding-space redundancy pruning

Stage 1 removes patches with no content. Stage 2 removes patches with duplicate content: the interior of a filled table cell, the middle of a thick rule, a run of uniform texture inside a photograph. These are salient to a pixel detector and yet contribute nothing to a MaxSim score. We visit the survivors in descending saliency order and greedily form clusters of vectors within cosine $\theta = 0.92$ of the representative, replacing each cluster with its renormalised mean.

This is close to free in ranking terms for a structural reason. MaxSim takes, per query vector, the maximum similarity over document vectors. If two document vectors lie within cosine θ of one another, the maximum over the pair and the similarity to their renormalised mean differ by an amount bounded by their angular spread, which is small by construction — so collapsing a tight cluster barely moves the score while removing one stored vector per duplicate. The argument does not extend to loose clusters, which is why θ is set high.

D. Stage 3: quantization and asymmetric scoring

A 128-dimensional float32 vector occupies 512 bytes; retaining only the sign of each dimension gives 128 bits, or 16 bytes — a flat 32× on everything the pruner allowed through. These vectors are L2-normalised with approximately zero-centred dimensions, so $\text{sign}(\mathbf{d})$ preserves the orthant of $\mathbf{d}$ and discards only its position within that orthant.

We score asymmetrically: the document side is binarised, the query side is not. A fully symmetric scheme reduces MaxSim to a popcount Hamming distance and is cheaper still, but it discards the query magnitudes — and those are precisely the part that can be afforded, since a query contributes on the order of twenty vectors against a

corpus of millions. Asymmetric scoring costs one unpack and one float matrix product and recovers most of the ranking quality; the symmetric variant is retained only for comparison. For a query matrix Q and a page’s code matrix C ,

$$\text{MaxSim}(Q, C) = \sum_i \max_j \mathbf{q}_i \cdot \text{sign}(\mathbf{d}_j). \quad (2)$$

We also evaluate an int8 codec at $4\times$. Because the components of an L2-normalised 128-dimensional vector concentrate near $1/\sqrt{d}$ — the observed mean absolute component is 0.071 and the observed maximum 0.363 — the textbook $\text{round}(127v)$ mapping, which assumes the data spans $[-1, 1]$, leaves roughly two thirds of the available levels permanently unused. Rescaling by the range the data actually occupies costs no additional storage and reduces the mean cosine error on real vectors from 3.28×10^{-4} to 8.20×10^{-5} .

E. Memory accounting

One implementation detail is load-bearing for the claims in this paper. A naive scorer decodes the packed index into a float32 matrix before computing similarities, which restores the full $d \times 4$ bytes per vector at query time and returns the entire saving — an index of 2.82 KB per page on disk demanded 90.12 KB per page in RAM to search. The storage claim was true while the system-level claim behind it was not. We therefore score in blocks sized against a memory budget, releasing each block before the next is allocated; peak resident memory then tracks the budget (measured at $0.93\times$ of it) and is independent of corpus size, while producing scores bit-identical to the cached path. All index sizes reported below are computed from the pipeline’s own byte accounting rather than from files on disk, so they are neither inflated by container overhead nor deflated by filesystem compression.

IV. Experimental Setup

A. Configuration

Three experiments share one pipeline, one measurement protocol and one code path, differing only in encoder and corpus. We label them E1, E2 and E3 throughout. E1 and E2 differ in both variables at once; E3 exists to separate them, and forms the fourth cell of a 2×2 over {small, reference} encoders and {generated, real} pages.

E1: small encoder, generated corpus. The encoder is vidore/colSmol-256M (ColIdefics3), float32, on CPU; hardware is a laptop with 8 GB of DDR4 and no GPU. The corpus is 60 generated pages (30 documents of 2 pages) rendered at A4, 150 dpi. The encoder emits 875 vectors per page before compression (768 page-grid patches, 64 thumbnail, 43 instruction).

E2: reference encoder, ViDoRe. The encoder is ColPali-v1.3 (3B parameters) in bfloat16 on a single Tesla T4. The corpus is four ViDoRe test splits — docvqa (500 pages, 451 queries), syntheticDocQA energy (500, 100), infovqa

(500, 494) and tabfquad (280, 280) — for 1,780 pages and 1,325 queries in total, with the benchmark’s own ground truth. The encoder emits 1,031 vectors per page.

E3: reference encoder, generated corpus. ColPali-v1.3 as in E2, over E1’s corpus. The generator is deterministic, so regenerating at seed 7 reproduces the pages E1 measured rather than pages resembling them; we verified that the manifests match. The encoder emits 1,031 vectors per page, as in E2. E1 and E3 therefore share a corpus, E2 and E3 share an encoder, and each comparison isolates one variable.

Vector dimension is 128 in all three and every run uses seed 1337.

One implementation detail is worth recording, because it silently produces plausible numbers. ColPali is published as a LoRA adapter over a PaliGemma base. On current transformers and PEFT releases the adapter’s key prefixes do not match the names of the injected modules, so the projection head is created and then left randomly initialised; the loader reports it as a missing key rather than as an error. A benchmark run in that state completes normally and yields a complete, well-formed, meaningless table. We therefore load the merged-weight checkpoint, which contains no adapter to mis-key, and our loader refuses to run if any parameter remains unmaterialised after loading.

B. Queries and ground truth

E1 has 72 queries in two deliberately different families. Sixty precise queries pair a subject with a unique identifier and have exactly one relevant page; these check that compression has not broken retrieval outright. Twelve topical queries name only a subject shared by several pages, so the metric depends on ordering a group of near-identical candidates — which is where compression damage becomes visible. Ground truth is exact by construction.

E2 uses the queries and relevance judgements shipped with each ViDoRe split, so the query distribution is the benchmark’s rather than ours. The precise/topical distinction is a property of our generator and does not carry over, so E2 is reported per split instead.

C. Controls

Two choices make the ablation a controlled experiment rather than a leaderboard.

Exact scoring. Every quality figure comes from exhaustive brute-force MaxSim, not from an approximate index. An ANN structure would contribute its own recall loss and confound the attribution we are trying to make. The Qdrant backend is tested separately and is the deployment path, not the measurement path.

One encoding pass. The corpus is encoded once into a cache and every ablation row replays over those identical vectors. Nothing but the compression setting differs between rows, so a difference in a metric can have no other

TABLE I

E1. Ablation over the compression pipeline. 60 pages, 72 queries, ColSmol-256M, exact MaxSim. Rows share one encoding pass, so every difference is attributable to the compression setting alone. Retain is nDCG@5 relative to the baseline row; τ is Kendall τ -b against the baseline’s own ranking.

Variant	What it isolates	Tok/pg	KB/pg	Compr.	nDCG@5	R@1	Hit@5	Retain	τ
baseline-float32	reference: every patch, full precision	875.0	448.00	1.0×	0.7823	0.5694	0.9722	100.0%	1.000
Pruning alone (no quantization)									
spatial-only	blank-patch pruning	356.1	182.32	2.5×	0.7602	0.5694	0.9444	97.2%	0.935
spatial+redundancy	both pruning stages	246.8	126.34	3.5×	0.7519	0.5139	0.9722	96.1%	0.866
Quantization alone (no pruning)									
int8-only	scalar quantization	875.0	112.00	4.0×	0.7877	0.5694	0.9861	100.7%	0.972
binary-only	one-bit quantization	875.0	14.00	32.0×	0.6875	0.4167	0.9444	87.9%	0.585
Both stages									
prune+int8	quality-first operating point	246.8	31.59	14.2×	0.7511	0.5139	0.9722	96.0%	0.864
optivision	size-first operating point	246.8	3.95	113.5×	0.6782	0.4028	0.9583	86.7%	0.606
optivision-aggressive	fixed 25% token budget	186.3	2.98	150.3×	0.6680	0.3611	0.9722	85.4%	0.602
Token-budget sweep, all with one-bit codes									
keep-50%	top 50% by saliency	288.1	4.61	97.2×	0.6704	0.4167	0.9444	85.7%	0.548
keep-40%	top 40% by saliency	263.6	4.22	106.2×	0.6721	0.4167	0.9583	85.9%	0.561
keep-30%	top 30% by saliency	241.6	3.87	115.9×	0.6721	0.4028	0.9583	85.9%	0.588
keep-20%	top 20% by saliency	206.1	3.30	135.9×	0.6676	0.3750	0.9722	85.3%	0.586
keep-10%	top 10% by saliency	162.9	2.61	171.8×	0.6700	0.4167	0.9306	85.6%	0.551

TABLE II

E2. The same ablation under ColPali-v1.3 on the ViDoRe docvqa split (500 pages, 451 queries), exact MaxSim, one encoding pass shared by every row. Compare with Table I: the binary rows have moved from the mid-80s to the mid-90s, and the pruning rows have lost most of their compression.

Variant	What it isolates	Tok/pg	KB/pg	Compr.	nDCG@5	R@1	Hit@5	Retain	τ
baseline-float32	reference: every patch, full precision	1031.0	527.87	1.0×	0.5841	0.4945	0.6608	100.0%	1.000
Pruning alone (no quantization)									
spatial-only	blank-patch pruning	846.3	433.30	1.2×	0.5794	0.4945	0.6519	99.2%	0.873
spatial+redundancy	both pruning stages	557.5	285.43	1.8×	0.5691	0.4856	0.6386	97.4%	0.715
Quantization alone (no pruning)									
int8-only	scalar quantization	1031.0	131.97	4.0×	0.5838	0.4989	0.6563	99.9%	0.919
binary-only	one-bit quantization	1031.0	16.50	32.0×	0.5625	0.4789	0.6364	96.3%	0.527
Both stages									
prune+int8	quality-first operating point	557.5	71.36	7.4×	0.5690	0.4878	0.6386	97.4%	0.712
optivision	size-first operating point	557.5	8.92	59.2×	0.5523	0.4545	0.6341	94.6%	0.510
optivision-aggressive	fixed 25% token budget	156.1	2.50	211.3×	0.4874	0.4080	0.5610	83.4%	0.409
Token-budget sweep, all with one-bit codes									
keep-50%	top 50% by saliency	373.9	5.98	88.2×	0.5404	0.4612	0.6142	92.5%	0.486
keep-40%	top 40% by saliency	309.0	4.94	106.8×	0.5276	0.4501	0.5987	90.3%	0.459
keep-30%	top 30% by saliency	239.8	3.84	137.6×	0.5076	0.4146	0.5942	86.9%	0.455
keep-20%	top 20% by saliency	165.3	2.64	199.6×	0.4778	0.3925	0.5543	81.8%	0.398
keep-10%	top 10% by saliency	86.0	1.38	383.6×	0.3899	0.3126	0.4634	66.7%	0.393

cause. Re-running the benchmark reproduces the reported figures exactly; the pipeline is deterministic given a fixed corpus and encoder.

D. Metrics

We report nDCG@5 with binary relevance (the standard ViDoRe metric), Recall@1 and Hit@5, index bytes per page, and the compression ratio against the uncompressed float32 index. We additionally report retention — nDCG@5 as a fraction of the baseline row — and Kendall τ -b against the baseline’s own ranking. The last is the load-bearing metric here. Absolute quality answers

“is this retriever any good?”, which depends mostly on the choice of encoder; τ answers “did compression change the retriever’s mind?”, which is the only question this layer controls.

V. Results

Table I is the full E1 ablation and Fig. 3 plots it. Table II is the same ablation under E2 on the split with the most queries, and Table III gives retention across all four. Three findings follow. Each is stated as E1 supports it and then tested against E2, and only the first survives in the form E1 suggested.

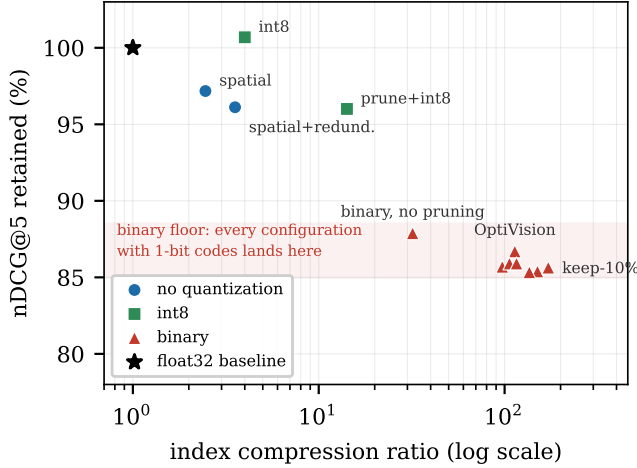


Fig. 3. E1. Compression ratio against nDCG@5 retention, by quantizer. The points separate by codec, not by token count: everything with one-bit codes falls into the shaded band between 85% and 88% retention whether it prunes nothing or prunes to a tenth, while the unquantized and int8 configurations stay above 96%. Fig. 4 shows what happens to this picture under E2.

TABLE III

E2. nDCG@5 retained across all four ViDoRe splits. The energy split has only 100 queries and reads above 100% for most rows; those differences are noise, not improvement. Compression ratios are rounded to whole numbers above 50 $\times$.

Variant	docvqa	energy	infovqa	tabfquad	Compr.
spatial-only	99.2%	101.1%	99.8%	99.8%	1.0–1.4 $\times$
spatial+redundancy	97.4%	101.8%	99.7%	99.2%	1.7–1.9 $\times$
int8-only	99.9%	101.6%	100.1%	99.9%	4.0 $\times$
binary-only	96.3%	100.6%	97.4%	100.3%	32.0 $\times$
prune+int8	97.4%	101.6%	99.5%	99.8%	6.7–7.5 $\times$
optivision	94.6%	103.4%	97.2%	95.6%	53–60 $\times$
keep-50%	92.5%	101.8%	94.7%	97.4%	81–91 $\times$
keep-10%	66.7%	89.8%	85.4%	81.6%	342–387 $\times$

A. Finding 1: pruning is close to free, and buys less than it appears

Under E1, removing blank patches costs 2.8% of nDCG@5 for a 2.5 $\times$ token reduction, and adding redundancy collapsing brings this to 3.5 $\times$ for 3.9%. Rank agreement with the uncompressed baseline remains high throughout ($\tau = 0.87$ –0.94), and spatial-only matches the baseline exactly on Recall@1 (0.5694): discarding 59% of the vectors does not cost a single top-1 hit.

Under E2 the quality claim holds more strongly and matters much less. Both pruning stages together retain 97.4–101.8% of nDCG@5 across the four splits, but the token reduction collapses from 3.5 $\times$ to 1.7–1.9 $\times$. The reason is visible in the extreme case: on infovqa, spatial pruning keeps 1,000 of 1,031 patches and saves essentially nothing. Wide margins are a property of the documents our generator renders, not of documents in general, and ViDoRe — scans, infographics, dense tables — does not have them. The claim survives; the number behind it does

not.

B. Finding 2: the token budget is free only when the corpus is sparse

Under E1 the token-budget sweep is almost flat. Keeping the top 10% of patches — 163 vectors per page — retains 85.6% of nDCG@5, while keeping the top 50% — 288 vectors, a 1.8 $\times$ larger index — retains 85.7%. The 0.1 point separating them is well inside the noise of 72 queries, which invites the conclusion that the token budget is not the operative variable once a codec is fixed.

Under E2 that conclusion fails outright. Keeping the top 10% retains 66.7% on docvqa, 81.6% on tabfquad and 85.4% on infovqa, against 92.5%, 97.4% and 94.7% at the top 50% (Table III). The flat sweep was an artifact of a corpus whose salient content fits comfortably inside the top 10% of patches; on pages where it does not, the aggressive setting is a genuine trade rather than a free 1.8 $\times$.

C. Finding 3: the one-bit penalty follows the decision margin, not the encoder

Under E1, binary-only loses 12.1% of nDCG@5 without discarding a single token, and its τ of 0.585 says it substantially reorders results rather than merely rescaling scores. Against this, int8-only costs nothing measurable at 4 $\times$ — 100.7% retention with $\tau = 0.972$, the excess above 100% being noise at 72 queries. Every E1 configuration that includes one-bit codes lands in a narrow 85–88% retention band regardless of how much pruning is applied. Under E2 the band is gone: binary-only retains 96.3% on docvqa and 97.4% on infovqa. Under E3 — ColPali over E1’s own pages — it retains 98.4% with R@1 unchanged at 0.375. A 12.1% penalty became 1.6% under a change of encoder alone, with the codec and the pages untouched (Table V).

The reading that suggests itself — that the codec’s cost is a property of the encoder — does not survive a per-query look at the same caches. For each query we take the float-precision margin m by which the gold page beats its best non-relevant competitor, as a fraction of the gold score, and the noise σ a codec adds to page scores, measured as the residual of the codec’s scores around a linear fit to the float scores in the same units. A sign code adds $\sigma \approx 2.8\%$ of the score under ColSmol and 2.5% under ColPali; a two-bit code about half that; int8 0.06%. The noise is a property of the codec, and it is the same for both encoders. What differs is where the queries sit. E1’s 60 precise queries are decided by three digits inside a nine-character code, and the retriever resolves them by a median margin of 0.05% (interquartile -0.7% to $+0.9\%$): the gold page is effectively tied with its same-subject neighbour before any codec is applied. The 12 topical queries are won by 31%. Under every codec, on every cache we hold (six ColSmol, two ColPali), no query whose margin exceeds 2σ changes its top result, while 32–43% of

TABLE IV

Per-query decision margin against codec noise, both as a fraction of the gold page’s float score. m is the median float margin of the 60 precise queries; σ is the score noise a codec adds; flip rates are the share of queries whose top-1 result changes under the sign code, by $|m|/\sigma$ bin; AUC is how well m alone separates the float winners the sign code keeps from those it loses. Rows are the E1 encoder (ColSmol) and the E3 encoder (ColPali) on the same 60 pages with the code’s glyphs at the stated size; the ColSmol rows re-render only those glyphs, the ColPali 3 $\times$ row was rendered with the code in the text flow, which also shifts the lines below it.

encoder, code	m	σ_{sign}	$\sigma_{2\text{bit}}$	$< 0.5\sigma$	$> 2\sigma$	AUC
ColSmol 1 $\times$ (E1)	−0.05%	2.8%	1.5%	32%	0%	0.78
ColSmol 3 $\times$	−0.31%	2.7%	1.4%	32%	0%	0.90
ColSmol 0.4 $\times$	−0.65%	2.8%	1.6%	41%	0%	0.94
ColPali 1 $\times$ (E3)	−1.39%	2.5%	1.3%	43%	0%	0.86
ColPali 3 $\times$	−0.52%	2.2%	1.2%	40%	0%	0.88

those below 0.5σ do; among the queries the float retriever wins, the margin alone separates the ones a sign code keeps from the ones it loses with AUC 0.78–0.95 (Table IV).

This is also why ColPali appears immune. On E1’s pages it wins 15 of the 60 precise queries in float, three more than the 12 a subject-only ranker gets by chance, and its median precise-query margin is −1.4%: it cannot read an 11 pt code at 448 px and has lost those queries before any quantization. Under the sign code seven of them flip from won to lost and seven from lost to won, which nets to the “unchanged” R@1 of 0.375. Re-rendering only the code’s glyphs three times larger, with every other line, the generator’s random sequence and the queries held fixed, lifts ColPali’s precise R@1 to 0.317 and its one-bit cost from 1.6% to 4.1%, with precise R@1 under the sign code back at 0.233: the encoder starts winning near-tie queries and starts losing them to the codec. At 72 queries the aggregate change is inside the noise (paired over identical queries, −0.019 nDCG@5, 95% CI [−0.094, +0.058]), so we report it as a direction; the per-query structure in Table IV is the measurement. The same manipulation on ColSmol, at 0.4 $\times$, 1 $\times$ and 3 $\times$ the code size, moves the float retriever (precise R@1 0.283, 0.483, 0.333 — a 33 pt code spans three patches, and no single document vector then holds the string a query vector must match) but leaves the one-bit cost at 9.6%, 12.1% and 9.0%, with paired differences of +0.03 nDCG@5 and intervals through zero. Glyph size, like encoder identity, is not the variable.

The statement that survives is narrower than the one the aggregate numbers invite, and more useful: a codec costs the queries whose decision margin is smaller than its score noise. E1 is the worst case by construction — 40 of its 72 queries sit below 0.5σ for a sign code — E3 is a floor, and E2’s 2.6–3.7% measures how many ViDoRe queries are within about 3% of a tie. Pruning’s yield, by contrast, does answer to the corpus and only to the corpus: 4.2 $\times$ to 1.9 $\times$ under one encoder, 3.6 $\times$ to 4.2 $\times$ under one corpus. At reference scale neither stage binds: compression to roughly 57 $\times$ is nearly free, and the limit

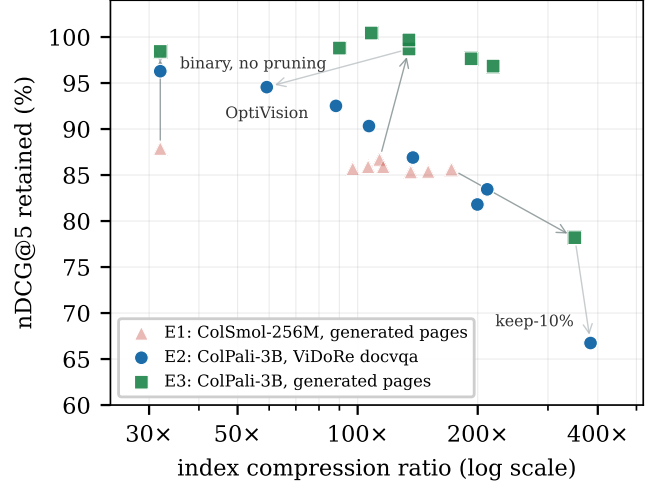


Fig. 4. The dissociation. Every configuration using one-bit codes, under all three experiments, at the same compression ratios. Under E1 (red) they sit in a band below 88% retention; under E2 (blue) the same configurations retain 94–97%. E3 (green) shares E1’s pages and E2’s encoder, and sits at or above E2, clear of E1’s band. Finding 3 shows why this is not an encoder property: E1’s band is made of near-tie queries that ColPali does not win in float precision either. Dark arrows join identical configurations from E1 to E3, which changes only the encoder; light arrows continue to E2, which changes only the corpus.

TABLE V

Each quantity against its own control. E3 shares E1’s corpus and E2’s encoder, so the first four rows change exactly one thing; the last two change nothing but the size of the one discriminative string on the page. Pruning’s yield answers to the corpus. The codec’s cost moves with the encoder and moves again with glyph size alone, and Finding 3 traces both to the queries’ decision margins. E2 figures are the docvqa split; the last two rows are at 72 queries and inside their confidence intervals.

Quantity	Control	Change
one-bit codec cost	corpus fixed, encoder swapped	12.1 $\rightarrow$ 1.6 pts
one-bit codec cost	encoder fixed, corpus swapped	1.6 $\rightarrow$ 3.7 pts
pruning yield	corpus fixed, encoder swapped	3.6 $\rightarrow$ 4.2 $\times$
pruning yield	encoder fixed, corpus swapped	4.2 $\rightarrow$ 1.9 $\times$
one-bit codec cost	ColPali, glyphs 3 $\times$	1.6 $\rightarrow$ 4.1 pts
one-bit codec cost	ColSmol, glyphs 0.4 $\times$ /3 $\times$	12.1 $\rightarrow$ 9.6/9.0

on the pipeline is how little a pixel-space saliency detector can remove from a dense page.

D. Finding 4: on dense pages, coverage beats ink and nothing beats random

Finding 2 leaves a question the pixel detector cannot answer: when the page is dense and a token budget is imposed anyway, which tokens should go? The natural alternative to ranking patches by ink is to rank them by how much of the embedding space they cover — score every document vector against a set of probe directions and keep the ones that win a MaxSim most often, so that whatever a query turns out to ask, some retained vector is near it. We built this selector with 256 probes chosen by

TABLE VI

Token selection on dense pages: retention of float nDCG@5 at three budgets, all rows one-bit. pixel is the saliency detector of Stage 1; coverage keeps the patches that win most MaxSims against 256 farthest-point probes; random is the same procedure with 256 random unit vectors. Budgets match to within 3% of tokens per page; 494 queries on infovqa, 451 on docvqa.

split	budget	pixel	coverage	random
infvqa	50%	94.7%	96.2%	96.7%
	30%	92.3%	95.9%	95.7%
	10%	85.4%	89.6%	90.7%
docvqa	50%	92.5%	91.4%	92.3%
	30%	86.9%	88.2%	90.5%
	10%	66.7%	77.9%	76.9%

greedy farthest-point traversal of a 20,000-patch sample of the corpus, and ran it on infvqa and docvqa at three budgets against the two controls any such selector has to beat: the same procedure with 256 uniformly random unit vectors that have never seen the corpus, and the pixel detector at the same budget (Table VI). Budgets match to within 3% of tokens per page, and every row uses the one-bit codec.

Two things are true at once. Coverage-based selection beats pixel saliency below a 50% budget on both splits, by 4 points at 10% on infvqa and 11 on docvqa: on a dense page the ink ranking discards the vectors queries actually match. And the probe construction contributes nothing to that. Random probes are within one point of the designed probes at every budget on infvqa, ahead by one to two points at 50% and 30% on docvqa, and behind by one at 10%; across 945 queries no selector we tried beats random. This reproduces, for a different selector and a different encoder, the observation of Ma et al. [10] that random pruning outperforms saliency-based pruning for ColPali, and we read it as a statement about the corpus rather than about any method. On these splits 74% (infvqa) and 66% (docvqa) of all patches win at least one MaxSim over the query set; the set of winners fitted on half the queries covers 64% and 52% of the index and retains 100% and 99.7% of nDCG@5 on the other half. That is the ceiling for any query-agnostic selector on dense pages — 1.6–1.9× at no loss — and below it every selector falls off a cliff at a similar rate. A learned selector for this regime has to beat uniformly random dropping at a matched budget, and we would regard that as the minimum control for any future claim of the kind.

E. Two operating points

Table VII summarises the practical consequence, and it differs between the two regimes. Under E1 the choice is a real exchange: prune + binary buys an 8× smaller index than prune + int8 and pays 9.3 points of retention for it, and which side is correct is a deployment question.

Under E2 that exchange collapses. Prune + binary gives roughly 8× the compression of prune + int8 at

TABLE VII

The configurations that follow from each experiment. Under E1 the two are a genuine exchange; under E2 and E3 the size-first point dominates on nDCG@5, and only τ separates them. E3 is the most favourable point we measure, but on sparse generated pages rather than real ones.

	Goal	Configuration	Compr.	Retain
E1	smallest index	prune + binary	113.5×	86.7%
E1	quality per byte	prune + int8	14.2×	96.0%
E2	smallest index	prune + binary	53–60×	94.6–103.4%
E2	quality per byte	prune + int8	6.7–7.5×	97.4–101.6%
E3	smallest index	prune + binary	134.5×	98.7%
E3	quality per byte	prune + int8	16.8×	102.3%

the same nDCG@5, so on the headline metric the size-first configuration simply dominates. What still separates them is ordering: Kendall τ is 0.71–0.90 for prune + int8 against 0.51–0.70 for prune + binary. A pipeline whose output is a shortlist for a downstream reader should take the binary configuration and the 57×; one that shows a ranked list directly to a person should still take int8, and the justification is τ rather than nDCG@5. This is a narrower and more specific recommendation than E1 alone would have produced, and it is the opposite of the one E1 alone would have produced.

E3 makes the same recommendation more emphatically — 134.5× at 98.7% retention — but it should not be read as the headline. It combines the corpus with the most blank paper to discard with an encoder that never wins the near-tie queries a one-bit code would cost it (Finding 3), which is the best case rather than a representative one. Its role here is to identify which variable supplies which part of the gain.

F. Latency

Query latency falls with the token count, since MaxSim cost is linear in stored vectors: under E1 from 3.25 ms at the float32 baseline to 0.70 ms at keep-10% on 60 pages, and under E2 on the 500-page docvqa split from 55.9 ms to 23.5 ms for the full pipeline and 4.4 ms at keep-10%. We do not emphasise these figures: the exact index used for measurement is not the structure a deployment would use, and query encoding dominates search entirely at this scale — 65.7 ms under E1 and 72.8 ms under E2.

VI. Discussion

A. What the one-bit codec costs, and why

For random unit vectors the expected cosine between $\mathbf{d}$ and $\text{sign}(\mathbf{d})/\sqrt{d}$ is $\sqrt{2/\pi} \approx 0.798$, and it is tempting to read an encoder’s departure from that figure as its susceptibility to one-bit codes. We measured it on every cache we hold: 0.800 for ColSmol, and 0.800, 0.800, 0.799 and 0.801 for ColPali on four corpora, against the random reference of 0.798. The statistic is a per-vector property of any roughly isotropic 128-dimensional distribution and carries no information about retrieval. The same is true

of the share of per-query-vector argmax identities that change under the sign code, which is 75–83% on every cache, including those where the ranking barely moves. Neither quantity can be the mechanism, and we withdraw both from the argument.

What a codec does to a ranking is decided one query at a time by two numbers. The first is the noise it adds to a page’s score, σ , which for a sign code is 2.2–2.8% of the score across both encoders and all corpora, for the two-bit code about half that, and for int8 0.06%; it is set by the codec and, weakly, by how anisotropic the encoder’s vectors are on that corpus. The second is the margin m by which the query’s gold page leads its best competitor in float, which is set by the corpus and by what the retriever can read. A query flips when $|m| \lesssim \sigma$. On the generated corpus the precise queries are near-ties by construction, so a sign code flips 32–43% of those below 0.5σ and none above 2σ , under either encoder; ColPali escapes only because its margins on those queries are negative to begin with. On ViDoRe a sign code costs 2.6–3.7%, which is the fraction of real queries within about 3% of a tie, and it changes the top-ranked page of 11% of infovqa queries while nDCG@5 reports 97.4% retention — a reason to report top-1 agreement alongside τ .

The ladder of codecs follows. Int8 is lossless at $4\times$ on every cache (99.9–100.2%). A rotated two-bit Lloyd–Max code at $16\times$ halves σ and recovers about half the sign code’s loss everywhere (99.4% on infovqa, 97.0% on docvqa, 99.4% on E3). The zero-byte refinements of the sign code — subtracting a global mean, an ITQ rotation, or a ColBERTv2-style centroid with a one-bit residual [3] — recover 3–10 points on E1, where 80% of every page is blank paper and the mean vector is the paper’s direction, and nothing on ViDoRe: +0.7 (95% CI $[-0.2, +1.8]$) and +0.7 $[-0.6, +2.0]$ on infovqa for centring and ITQ, -0.8 $[-2.3, +0.7]$ and -0.2 $[-2.0, +1.6]$ on docvqa, and -4.2 $[-6.5, -2.1]$ for the residual code on docvqa with its codebook capped at 4,096 centroids for 515,500 vectors. The anisotropy those refinements correct is a property of blank pages, not of the encoder, and it is why E1 overstates their value.

B. Implications

Three follow. First, and most strongly, a reported compression ratio for multi-vector retrieval should carry both a per-stage attribution and the encoder and corpus it was measured on. Our first two experiments differ only in those two variables and produce opposite attributions from the same code; a paper reporting either one alone would have supported a confident and wrong generalisation, and it took a third run to establish which variable was responsible for which half of the difference.

Second, the token budget is a free parameter exactly when the corpus is sparse, and not otherwise. E1 would have licensed pruning as aggressively as the detector allows; on ViDoRe that costs 25.8 points of retention

between the top 50% and the top 10%. Sparsity is cheap to measure — it is the fraction of patches the saliency detector rejects — and should be reported alongside any token-reduction result.

Third, the research direction the attribution recommends is different in each regime, which is the practical point of making the attribution at all. Where queries are near-ties — exact identifiers, codes, serial numbers on otherwise similar pages — the codec binds, the sign code is the wrong choice, and a two-bit code at $16\times$ or int8 at $4\times$ is what a deployment should pay for. Where queries are decided by wide margins the codec is nearly free and the binding limit is the selector: on dense pages pixel saliency discards the vectors queries match, coverage-based selection does better below a 50% budget, and uniformly random probes do as well as any selector we tried (Finding 4). A learned selector for this regime has to beat random at a matched budget before it has earned anything. Shortlist rescoring remains attractive in both, since Hit@5 stays well above nDCG@5 throughout — the relevant page is usually retrieved and merely misordered.

C. Limitations

We state these plainly, as they bound the claims above.

Corpus scale. E2 uses 500-page subsets of each ViDoRe split rather than the full splits, so absolute nDCG@5 is not comparable with published leaderboard figures. Only the ratios between rows are the measurement here, and those are what we claim; but a larger corpus makes retrieval harder, and the compression penalties we report may be optimistic in absolute terms.

Statistical power. E1 has 72 queries and the energy split of E2 has 100, at which several rows read above 100% retention. Those readings are noise and we decline to interpret them; the docvqa (451) and infovqa (494) splits carry the E2 conclusions. Every E3 aggregate difference is likewise at 72 queries and inside ± 0.08 nDCG@5; the per-query margin structure of Table IV, not the aggregates, carries the E3 conclusion.

Two points, one family. We measure a 256M and a 3B encoder, both ColBERT-style vision–language retrievers. E3 separates encoder from corpus, but two points still do not locate a crossing: we cannot say at what scale the attribution changes, nor whether parameter count or embedding quality is the operative variable. The decision-margin rule of Finding 3 is verified on the generated corpus, eight caches deep, and is a prediction rather than a measurement on ViDoRe, whose caches we did not retain. The ColPali run with the enlarged code was rendered with the code in the text flow, which also shifts the lines below it; the three ColSmol runs hold every other pixel fixed.

Generator realism. On E3’s pages ColPali-v1.3 scores below ColSmol-256M (nDCG@5 0.6954 against 0.7823). A 3B reference encoder losing to a 256M one is a fact about our generated pages rather than about the encoders, and it bounds what E1 and E3 can be taken to represent. It

does not affect the comparisons we draw from them, which are within-encoder ratios.

What τ measures. Our τ is Kendall τ_b over the shared ids of two top-10 hit lists, not over the full ranking; the same E1 run scores 0.643 over all 60 pages, and the value moves with the cutoff (0.402 at top-5, 0.691 at top-20). It should therefore be read with its cutoff named, and τ values are comparable across experiments only when the candidate pool is a comparable fraction of the corpus — which holds for E1 against E3, both 60 pages, and not for either against E2’s 500. Where one number is wanted we now prefer top-1 agreement and top-5 overlap with the float ranking, which are free of a cutoff and report what a user would see; under the sign code on infovqa they are 0.89 and 0.77.

Scoring path. All quality numbers come from exact brute-force MaxSim, deliberately. A deployment on an approximate index would see additional recall loss from the ANN structure, which our figures do not include and were designed not to include.

VII. Conclusion

We built a model-agnostic compression layer for vision-language document retrieval and used it to ask which of its two stages the retrieval quality is actually paid to. Asking the question twice is what the paper turns on. Under a 256M encoder on generated pages the two stages are sharply asymmetric: pruning costs 3.9% of nDCG@5 for a $3.5\times$ token reduction while one-bit quantization costs 12.1% having discarded no tokens at all, and every binary configuration lands in the same 85–88% band. Under ColPali-v1.3 on four ViDoRe splits the same code produces the opposite attribution: one-bit quantization costs at most 3.7%, the full pipeline reaches 53–60 $\times$ at 94.6–103.4% retention, and what fails instead is the premise that a page is mostly blank paper — pruning yields 1.7–1.9 $\times$ on real pages, and a token-budget sweep that was flat becomes a 25.8-point drop.

A third experiment holds the corpus fixed and swaps only the encoder, and a per-query analysis of the caches separates what that reversal confounded. The corpus governs what pruning buys (4.2 $\times$ to 1.9 $\times$ under one encoder). The codec’s cost is not an encoder property: a sign code adds the same 2–3% of score noise under either encoder, and it costs exactly the queries whose float margin over the best competitor is smaller than that — none above twice the noise on any of eight caches. The generated corpus decides its precise queries at a median margin of 0.05%, which is why the small encoder looks fragile; the reference encoder never wins those queries in the first place, which is why it looks immune, and enlarging the code until it can lift its precise R@1 off the floor and its one-bit cost with it. On dense pages, selecting tokens by embedding coverage beats pixel saliency below a 50% budget and nothing beats uniformly random probes. The conclusion is therefore not that the codec binds, nor

that the token count binds, but that neither is a property of the compression layer alone. A compression ratio for multi-vector retrieval should be published with a per-stage attribution and with the encoder, corpus and query regime that attribution was measured on, because the attribution itself is what moves.

Future work follows directly: verify the decision-margin rule on the ViDoRe caches, where it is so far a prediction; place the two-bit code, which halves the sign code’s noise at twice its bytes, at the operating point for near-tie query regimes; hold any learned token selector to the random-dropping control at a matched budget; and test shortlist rescoring, which our Hit@5 figures suggest should recover most of the remaining ranking loss at negligible cost.

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